Introduction

The Bland-Altman plot is the standard graphical tool for assessing agreement between two measurement methods, two raters, or two devices. Unlike a correlation coefficient or a 45-degree scatter, which conflate agreement with linear association, a Bland-Altman plot displays the *difference* between paired measurements against their *mean*, exposing systematic bias, proportional bias, and the limits within which 95 % of differences are expected to fall. Since Bland and Altman's 1986 *Lancet* paper, it has become the canonical method-comparison figure in clinical chemistry and laboratory medicine.

Prerequisites

A working understanding of paired measurements, mean and standard deviation, and the distinction between agreement (do two methods give the same value?) and correlation (do two methods rank observations the same way?).

Theory

For paired measurements (X_i, Y_i), compute the mean $M_i = (X_i + Y_i)/2$ and the difference $D_i = X_i - Y_i$. Plot M on the x-axis and D on the y-axis, with three horizontal reference lines:

- **Bias**: $\bar{D}$, the mean of the differences. Non-zero bias indicates systematic disagreement.
- **Upper limit of agreement (LoA)**: $\bar{D} + 1.96 \cdot \mathrm{SD}(D)$.
- **Lower LoA**: $\bar{D} - 1.96 \cdot \mathrm{SD}(D)$.

If the differences are approximately Normal, about 95 % of points fall within the LoA. Trends in D as a function of M — typically tested by regressing D on M — indicate proportional bias, where the disagreement scales with magnitude. Heteroscedasticity (a fan-shaped pattern) suggests the LoA should be expressed as a percentage of the mean rather than as an absolute interval.

Assumptions

Differences approximately Normal, no proportional bias (constant spread across the measurement range), and independent paired observations. For repeated measures within subjects, the modified Bland-Altman 1999 method partitions within- and between-subject variance.

R Implementation

```r
library(ggplot2); library(blandr)

set.seed(2026)
m1 <- rnorm(80, 100, 15)
m2 <- m1 + rnorm(80, 2, 5)

d <- data.frame(M = (m1 + m2)/2, D = m1 - m2)
bias <- mean(d$D)
loa <- bias + c(-1, 1) * 1.96 * sd(d$D)

ggplot(d, aes(M, D)) +
  geom_point(alpha = 0.6) +
  geom_hline(yintercept = bias, colour = "#2A9D8F", linewidth = 1) +
  geom_hline(yintercept = loa, colour = "#F4A261", linetype = "dashed") +
  labs(x = "Mean of methods", y = "Difference (M1 - M2)") +
  theme_minimal()
```

```r
# With blandr
blandr.draw(m1, m2, plotter = "ggplot")
```

Output & Results

A scatter of differences vs means with three reference lines: a solid line at the bias of approximately 2 units and two dashed lines at the upper and lower limits of agreement of approximately -7.8 and 11.8 units. About 4 of the 80 points (5 %) should fall outside the LoA under the Normality assumption.

Interpretation

A reporting sentence: "Bland-Altman analysis showed a bias of +2.0 units (95 % CI 0.9 to 3.1) and 95 % limits of agreement of -7.8 to 11.8 units; clinical equivalence requires the LoA to fall within ±5 units, so the methods are not interchangeable as currently calibrated." Always report the LoA with their 95 % confidence intervals — the LoA endpoints are themselves uncertain, especially for $n < 100$.

Practical Tips

- Check for proportional bias by regressing D on M; a significant slope indicates the disagreement scales with magnitude and the constant LoA misrepresents the data.
- Report the LoA with confidence intervals; the standard formulae give intervals of $\pm 1.71 \cdot \mathrm{SE}$ around each LoA.
- For repeated measures within subjects, use Bland-Altman's 1999 modified method that splits the variance into within- and between-subject components; ignoring the structure underestimates the LoA.
- A 45-degree scatter complements (but does not replace) the Bland-Altman plot; it reveals correlation but hides the level-dependent bias the Bland-Altman exposes.
- For heteroscedastic differences, log-transform both measurements first; the resulting Bland-Altman is on the log scale and the LoA become percentages.
- `blandr::blandr.statistics()` returns the full statistical summary including bias, LoA, their CIs, and a Shapiro-Wilk test for the Normality of differences.

R Packages Used

`blandr` for `blandr.statistics()` and `blandr.draw()`; `BlandAltmanLeh` for an alternative interface; `ggplot2` for hand-built versions; `epiR::epi.ccc()` for the concordance correlation coefficient as a complementary single-number agreement summary.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- ggplot2 grammar and multi-panel layouts